

MINAS KATSIOKALIS

XR Software Engineer

 minaskatsiokalis.com  minaskatsiokalis@gmail.com
 +49 175 9636561  github.com/MinasKatsiokalis
 Bonn, Germany  in/minas-katsiokalis

SUMMARY

I am an XR Software Engineer, currently specializing in XR development for Meta Quest, Apple Vision Pro, and handheld, and previously on HoloLens, with several years of hands-on Unity and C#/.NET experience shipping immersive applications. I focus on building reusable, well-structured systems and frameworks, and I have experience integrating REST services into client applications within defined architectures. Comfortable making pragmatic decisions in fast-moving environments, I am driven to advance immersive technology and, in particular, to use it for effective hands-on learning and skills training.

SKILLS

Languages: C#/.NET, Python, JavaScript, HTML/CSS, C/C++
Engines: Unity3D
XR SDKs: AR Foundation (ARCore/ARKit), Vuforia, MRTK (OpenXR, OculusXR), Meta XR SDK, Lightship ARDK, Apple Polyspatial
Platforms: Quest 2/3/Pro, HoloLens2, Vision Pro, Android, iOS
Versioning: Git

EXPERIENCE

- 8/2024 – Present **XR Software Engineer** **ANIMA RES GmbH**
- Deliver and ship production releases (e.g. Insight Heart) on schedule, balancing pragmatic technical decisions with quality standards.
 - Implement new features and enhancements for the Insight app gallery, collaborating closely with 3D and technical artists to ensure high-quality visuals and optimal performance on HMD devices.
 - Maintain and extend the company's custom-built Unity package system, which standardizes development across multiple platforms (Apple Vision Pro, Meta Quest, HoloLens, Android, iOS).
 - Implemented a package to add Meta Quest as a supported target platform, and refactored the HMD packages for Apple Vision Pro, extending and improving the reusability of the company's custom-built package system.
 - Improved Networking solutions for multi-user experiences and extended networking systems to support SharePlay service (Apple's native multi-user solution)
 - Implemented the integration of a WebXR (8th Wall) solution into the company's development pipeline.
 - Integrated Claude Code into everyday development work, and structured projects to be AI-agent-friendly for optimised AI agent communication.
- Unity3D / C# / XRI Toolkit / OpenXR / AR Foundation / MRTK / Polyspatial / Android (ARCore) / iOS (ARKit) / WebXR (8th-Wall)
- 10/2021 – 7/2024 **AR Software Engineer** **Up2metric P.C.**
- Designed and implemented mobile AR applications for Android and iOS, from prototyping through to production features.
 - Built and maintained reusable AR features and components, troubleshooting and resolving complex AR-related issues.
 - Integrated client applications with backend services by consuming REST, GraphQL, and WebSocket endpoints, handling real-time data exchange between client and server.
 - Researched XR standards and specifications to inform technical decisions and ensure best practices.
- Unity3D / C# / AR Foundation / Vuforia / Android (ARCore) / iOS (ARKit)
- 9/2021 – 3/2024 **XR Software Engineer & Researcher** **Technical University of Crete**
- Designed and implemented MR HMD applications for Meta Quest and HoloLens 2.
 - Researched XR collaborative environments and multimodal communication and interaction techniques.
 - Prototyped and tested HMD features, iterating on interaction design and usability.
 - Consumed REST APIs to connect HMD client applications with backend services.
 - Participated in HORIZON EU research projects (SUN XR, VARCITIES).
- Unity3D / C# / MRTK / Meta XR SDK / Vuforia / HCI
- 2022,2023 Spring Semesters **Graphics Lab Assistant** **Technical University of Crete**
- As post-graduate student I contributed to my supervisor's courses:
- **Graphics Course (Undergraduate - 4th year):**
 - Delivered practical instruction in Unity3D for desktop game development.
 - Designed, prepared, and tested semester-long student projects using Unity3D.
 - **Human-Computer Interaction (HCI) Course (Postgraduate):**
 - Organized and guided the semester project, including use-case design, persona development, usability evaluation, UX design, prototyping, and web app implementation.
 - Contributed to the preparation, review, and evaluation of course materials and assignments.
- Unity3D / C# / HCI

9/2018 – 5/2020
Diploma Thesis

AR Researcher & Developer

Technical University of Crete

- Develop a Mobile AR application for Android devices.
- Integrate navigation system using Mapbox SDK.
- Research and testing on various AR SDKs: Vuforia, Wikitude, ARCore, ARKit etc.
- Research on the field of Mobile AR (MAR).

Unity3D / C# / AR Foundation / Vuforia / ARCore / Wikitude

EDUCATION

2020

MEng, Integrated Masters in Electrical and Computer Engineering (5 Year Course - 300 ECTS)

Technical University of Crete

Diploma Thesis: "Outdoors Mobile Augmented Reality for Coastal Erosion Visualization, based on Geographical Data" a Mobile AR application for Android devices, developed in Unity3D game engine.

PUBLICATIONS

11/2024

International Conference on Extended Reality, XR Salento 2024

Springer

Engineering a BIM-Based Mixed Reality Application for the Life-Cycle Management of Buildings

12/2023

Journal: Smart Cities 2024, 7(1), 1-32

MDPI

The Intersection of the Green and the Smart City: A Data Platform for Health and Well-Being through Nature-Based Solutions

6/2023

ACM International Conference on Interactive Media Experiences (IMX' 23)

ACM Library

GoNature AR: Air Quality & Noise Visualization Through a Multimodal and Interactive Augmented Reality Experience

6/2022

7th International Conference on Smart and Sustainable Technologies (SpliTech)

IEEE Xplore

On the integration of nature-based solutions with digital innovation for health and wellbeing in cities

11/2020

International Workshop on Cross-Reality (XR) Interaction @ ACM ISS 2020

Workshop Proceedings

Outdoors Mobile Augmented Reality for Coastal Erosion Visualization, based on Geographical Data

* *Google Scholar*.

AWARDS

6/2023

Best Demo Award

ACM International Conference on Interactive Media Experiences (IMX' 23)

Received "Best Demo Award" of the conference by Tobbi for the demo: "GoNature AR".

2/2023

Winner of "Innovation" Category

Brackeys GameJam 2023.1

Category winners for the game: "UnSentenced"

LANGUAGES

• **English** - Fluent • **German** - Beginner • **Greek** - Native